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Interconnect structure including topological material

Abstract

A semiconductor device includes a substrate and an interconnect layer disposed over the substrate. The interconnect layer includes an interconnect structure which includes a topological material. The topological material includes a topological insulator, a topological semimetal, or a combination thereof. A method for manufacturing the semiconductor device is also disclosed.

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Background/Summary

BACKGROUND

(1) The integration density of various electronic components, such as transistors, diodes, resistors, capacitors, etc., is being improved continuously in the semiconductor industry by continual reduction in minimum feature size. As the feature size decreases, the dimensions of interconnects (i.e., metal lines and/or via contacts) for interconnecting these electronic components are scaled down as well. However, such interconnects would suffer high resistance problems when the dimensions thereof are scaled down.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a flow diagram illustrating a method for manufacturing a semiconductor device in accordance with some embodiments.

(3) FIGS. 2 to 12 are schematic views illustrating some intermediate stages of the manufacturing

method as depicted in FIG. 1 in accordance with some embodiments.

(4) FIG. 13 is a schematic view illustrating a semiconductor device in accordance with some alternative embodiments.

(5) FIG. 14 is a schematic view illustrating a semiconductor device in accordance with some alternative embodiments.

(6) FIG. 15 is a schematic view illustrating a semiconductor device in accordance with some alternative embodiments.

(7) FIG. 16 is a schematic view illustrating a semiconductor device in accordance with some alternative embodiments.

DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(9) Further, spatially relative terms, such as “on,” “over,” “below,” “upper,” “lower,” “downwardly,” “upwardly,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(10) The present disclosure is directed to a semiconductor device including an interconnect layer formed with an interconnect structure including a topological material, and a method for manufacturing the same. FIG. 1 illustrates a method **100** for manufacturing a semiconductor device in accordance with some embodiments. FIGS. 2 to 12 are schematic views of a semiconductor device **200** at some intermediate stages of the manufacturing method as depicted in FIG. 1 in accordance with some embodiments. Additional steps can be provided before, after or during the method **100**, and some of the steps described herein may be replaced by other steps or be eliminated. Similarly, further additional features may be present in the semiconductor device **200**, and/or features present may be replaced or eliminated in additional embodiments.

(11) Referring to FIG. 1 and the example illustrated in FIG. 2, the method **100** begins at step **101**, where a topological material layer is formed on an inter-layer dielectric layer. FIG. 2 is a schematic view illustrating formation of a topological material layer **12** on an inter-layer dielectric (ILD) layer **11** disposed over a substrate **10**.

(12) In some embodiments, the substrate **10** is a semiconductor substrate which may include, for example, but not limited to, an elemental semiconductor or a compound semiconductor. An elemental semiconductor includes a single species of atoms, such as silicon (Si) or germanium (Ge) in column XIV of the periodic table, and may be in crystal, polycrystalline, or an amorphous form. Other suitable materials are within the contemplated scope of the present disclosure. A compound semiconductor includes two or more elements, and examples thereof may include, but not limited to, silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), indium antimonide (InSb), silicon germanium (SiGe), gallium arsenide phosphide (GaAsP), aluminum indium arsenide (AlInAs), aluminum gallium arsenide (AlGaAs), gallium indium arsenide (GaInAs), gallium indium phosphide (GaInP), and gallium

indium arsenide phosphide (GaInAsP). Other suitable materials are within the contemplated scope of the present disclosure. The compound semiconductor may have a gradient feature in which the compositional ratio thereof changes from one location to another location therein. The compound semiconductor may be formed over a silicon substrate. The compound semiconductor may be strained. In some embodiments, the semiconductor substrate may include a multilayer compound semiconductor structure. In some embodiments, the semiconductor substrate may be a semiconductor on insulator (SOI) (e.g., silicon germanium on insulator (SGOI)). Generally, an SOI substrate includes a layer of a semiconductor material such as epitaxial silicon, germanium, silicon germanium, or combinations thereof. Other suitable materials are within the contemplated scope of the present disclosure. The SOI substrate may be doped with a P-type dopant, for example, but not limited to, boron (Br), aluminum (Al), or gallium (Ga). Other suitable materials are within the contemplated scope of the present disclosure. Alternatively, the SOI substrate may be doped with an N-type dopant, for example, but not limited to, nitrogen (N), phosphorous (P), or arsenic (As). Other suitable materials are within the contemplated scope of the present disclosure. In some embodiments, the semiconductor substrate may further include various active regions, for example, the active regions configured for an N-type metal oxide semiconductor transistor device (NMOS) or the active regions configured for a P-type metal oxide semiconductor transistor device (PMOS).

(13) In some embodiments, the ILD layer **11** may include dielectric materials, such as silicon oxide (SiO_x), silicon nitride (SiN_x), silicon carbide (SiC_x), silicon oxycarbide (SiO_xCy), silicon oxynitride (SiO_xNy), hydrogenated silicon oxycarbide (SiO_xCyH_z), spin-on glass (SOG), amorphous fluorinated carbon, fluorinated silica glass (FSG), Xerogel, Aerogel, polyimide, Parylene, BCB (bis-benzocyclobutenes), Flare, SiLK (Dow Chemical Co., Midland, Mich.), non-porous materials, porous materials, other low-k dielectric materials, or combinations thereof. Other suitable materials are within the contemplated scope of the present disclosure. In some embodiments, the ILD layer **11** may have a k-value ranging from about 1 to about 5. In some embodiments, the ILD layer **11** may be formed by a suitable deposition process known to those skilled in the art of semiconductor fabrication, for example, but not limited to, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), high density plasma CVD (HDPCVD), metal organic CVD (MOCVD), remote plasma CVD (RPCVD), plasma enhanced CVD (PECVD), or low-pressure chemical vapor deposition (LPCVD). Other suitable techniques for forming the ILD layer **11** are within the contemplated scope of the present disclosure. In some embodiments, the ILD layer **11** may be formed with at least one interconnect (e.g., a via contact, not shown) for interconnecting an interconnect structure to be formed from the topological material layer **12** with a conductive feature (not shown) disposed below the ILD layer **11**.

(14) In some embodiments, the topological material layer **12** may include a topological insulator, a topological semimetal, or a combination thereof. The topological insulator is a material which behaves as an insulator in its bulk, but which has a conductive state at the surface thereof, meaning that electrons can only move along the surface of the topological material. In some embodiments, the topological insulator may include, for example, but not limited to, bismuth telluride (Bi.sub.2Te.sub.3), antimony telluride (Sb.sub.2Te.sub.3), bismuth-antimony alloys (Bi.sub.1-xSb.sub.x), bismuth selenide (Bi.sub.2Se.sub.3), other bismuth (Bi)-based chalcogenides possessing a bulk band gap and a conductive surface state, or the like, or combinations thereof. Other suitable materials are within the contemplated scope of the present disclosure. The topological semimetal has a phase of matter analogous to the topological insulator but with non-insulating band structures, and is a material which behaves as a conductor in its bulk and which has a conductive state at the surface thereof as well. In some embodiments, the topological semimetal may include, for example, but not limited to, niobium phosphide (NbP), tantalum arsenide (TaAs), niobium arsenide (NbAs), tantalum phosphide (TaP), tungsten ditelluride (WTe.sub.2), molybdenum ditelluride (MoTe.sub.2), tungsten diphosphide (WP.sub.2), molybdenum

diphosphide (MoP.sub.2), cadmium arsenide (Cd.sub.3As.sub.2), platinum stannide (PtSn.sub.4), lanthanum antimonide (LaSb), lanthanum bismuthide (LaBi), platinum bismuthide (PtBi.sub.2), zirconium pentatelluride (ZrTe.sub.5), hafnium pentatelluride (HfTe.sub.5), lead tantalum selenide (PbTaSe.sub.2), zirconium silicon sulfide (ZrSiS), hafnium silicide sulfide (HfSiS), niobium arsenide (NbAs.sub.2), tantalum arsenide (TaAs.sub.2), other material systems possessing topological protected conductive surface states, or the like, or combinations thereof. Other suitable materials are within the contemplated scope of the present disclosure. In some embodiments, the topological material may be doped with a magnetic material to improve the transport properties of an interconnect structure to be formed from the topological material layer **12**. In some embodiments, the magnetic material may include, for example, but not limited to, cobalt (Co), iron (Fe), nickel (Ni), or the like, or combinations thereof. Other suitable materials are within the contemplated scope of the present disclosure. The topological material layer **12** may be formed on the ILD layer **11** by a suitable deposition process known to those skilled in the art of semiconductor fabrication, for example, but not limited to, molecular beam epitaxy (MBE) deposition, PVD, MOCVD, CVD, ALD, plasma-enhanced ALD (PEALD), PECVD, or the like. Other suitable techniques for forming the topological material layer **12** are within the contemplated scope of the present disclosure. In some embodiments, the deposition process for forming the topological material layer **12** may be performed at a temperature ranging from about 25° C. to about 1000° C. If the temperature for the deposition process is higher than 1000° C., the materials and the components disposed below the topological material layer **12** may be damaged. The crystallinity of the topological material layer **12** may be single crystal, near single crystal (i.e., the topological material layer **12** has a lateral grain size much greater than a thickness thereof), highly oriented polycrystal, or polycrystal. In some embodiments, the topological material layer **12** may be subjected to an annealing process to improve the crystallinity thereof and to reduce resistivity of the interconnect structure to be formed from the topological material layer **12**. In some embodiments, the annealing process may be, for example, but not limited to, a rapid thermal annealing (RTA) process, a laser process, a furnace annealing process, or the like. Other suitable annealing techniques are within the contemplated scope of the disclosure.

(15) Referring to FIG. **1** and the example illustrated in FIG. **3**, the method **100** then proceeds to step **102**, where the topological material layer is patterned to form a plurality of trenches. A mask layer **13** (for example, a hard mask layer) is deposited on the topological material layer **12**. Examples of a material suitable for forming the mask layer **13** include, for example, but not limited to, silicon oxide, silicon nitride, silicon carbide, silicon oxycarbide, silicon oxynitride, silicon carbonitride, silicon oxycarbonitride, tungsten nitride, tungsten carbide, titanium nitride, zirconium oxide, aluminum oxide, yttrium oxide, aluminum oxynitride, hafnium oxide, hafnium zirconium oxide, hafnium silicon oxide, hafnium silicon oxynitride, zirconium silicon oxide, hafnium zirconium silicon oxide, hafnium aluminum oxide, hafnium aluminum nitride, zirconium aluminum oxide, yttrium oxide, and combinations thereof. Other suitable materials are within the contemplated scope of the present disclosure. The mask layer **13** may be formed on the topological material layer **12** by a suitable deposition process as is known in the art of semiconductor fabrication, such as PVD, CVD, ALD, PEALD, thermal ALD, PECVD, or the like. Other suitable techniques are within the contemplated scope of the present disclosure. In some embodiments, the mask layer **13** has a thickness ranging from about 1 nanometer (nm) to about 1000 nm. If the thickness is less than 1 nm, the mask layer **13** may not be formed as a continuous film. If the thickness is greater than 1000 nm, the cost for forming the mask layer **13** is increased undesirably. A dielectric layer **14** may be optionally formed on the mask layer **13** to improve the flatness of the mask layer **13**. The material and the process for forming the dielectric layer **14** may be the same as or similar to those for forming the ILD layer **11** described above with reference to FIG. **2**, and the details thereof are omitted for the sake of brevity. A photoresist layer (not shown) is then formed on the dielectric layer **14** by a suitable fabrication technique known to those skilled in the art of

semiconductor fabrication, for example, but not limited to, a spin-on technique. Other suitable techniques are within the contemplated scope of the present disclosure. The photoresist layer is then patterned using a suitable photolithography technique to form a pattern of recesses. For example, the photoresist layer is exposed to light for patterning, followed by developing to form the pattern of the recesses. The pattern of the recesses formed in the photoresist layer is transferred to the dielectric layer **14** and then to the mask layer **13** using one or more etching processes, for example, but not limited to, a wet etching process, a dry etching process, a reactive ion etching process, a neutral beam etching process, or the like. After the pattern of the recesses is transferred to the mask layer **13**, the photoresist layer may be removed by, for example, but not limited to, an ashing process. The pattern of the recesses formed in the mask layer **13** is then transferred to the topological material layer **12** using a suitable etching process, for example, but not limited to, a plasma etching process, a sputter etching process, a reactive ion etching process, a deep-reactive ion etching process, or the like, so as to pattern the topological material layer **12** and to form a plurality of trenches **15** extending through the dielectric layer **14**, the mask layer **13**, and the topological material layer **12** to terminate at the ILD layer **11**. Other suitable etching techniques are within the contemplated scope of the present disclosure.

(16) Referring to FIG. **1** and the example illustrated in FIG. **4**, the method **100** then proceeds to step **103**, where a dielectric liner layer is conformally formed. A dielectric liner layer **16** is conformally formed to cover the dielectric layer **14**, the mask layer **13**, the topological material layer **12**, and the ILD layer **11** by a suitable deposition process as is known in the art of semiconductor fabrication, such as PVD, CVD, ALD, PEALD, PECVD, or the like. Other suitable techniques are within the contemplated scope of the present disclosure. In some embodiments, the dielectric liner layer **16** may include silicon oxide (SiOx), silicon carbide (SiCx), silicon nitride (SiNx), silicon oxycarbide (SiOxCy), silicon oxynitride (SiOxNy), aluminum oxide (AlOx), aluminum nitride (AlN), transition metal carbides, transition metal nitrides, transition metal oxides, or the like, or combinations thereof. Other suitable dielectric materials are within the contemplated scope of the present disclosure. In some embodiments, the dielectric liner layer **16** may have a thickness ranging from about 1 nm to about 10 nm. If the thickness is less than 1 nm, the liner/barrier properties of the dielectric liner layer **16** may be degraded. If the thickness is greater than 10 nm, the etched volume of the topological material layer **12** should be undesirably increased to provide each of the trenches **15** with a remaining volume sufficient for subsequent processing, such that the conduction volume of the interconnect structure formed from the topological material layer **12** is decreased and the resistance of the interconnect structure is increased.

(17) Referring to FIG. **1** and the examples illustrated in FIGS. **4** and **5**, the method **100** then proceeds to step **104**, where a dielectric layer is formed to fill the trenches. A dielectric layer **17** is formed by depositing a dielectric material on the dielectric liner layer **16**, so as to fill the trenches **15**. In some embodiments, the dielectric material and the deposition process for forming the dielectric layer **17** may be the same as or similar to those for forming the ILD layer **11** described above with reference to FIG. **2**, and the details thereof are omitted for the sake of brevity. A plurality of air gaps **18** are formed accordingly as a result of incomplete filling of the dielectric layer **17** into the trenches **15**.

(18) Referring to FIG. **1** and the examples illustrated in FIGS. **5** and **6**, the method **100** then proceeds to step **105**, where a first interconnect layer is formed. A suitable planarization process, for example, but not limited to, chemical mechanical planarization (CMP) is performed to remove a portion of the dielectric layer **17**, a portion of the dielectric liner layer **16**, the dielectric layer **14**, the mask layer **13**, and optionally a portion of the topological material layer **12** to form a first interconnect layer **19** disposed on the ILD layer **11**. The first interconnect layer **19** includes an interconnect structure **191**, a capping layer **192**, and a plurality of dielectric liners **193**. The interconnect structure **191** is formed from the topological material layer **12**, and includes a plurality of conductive lines **1911** spaced apart from each other by the air gaps **18**, such that the air gaps **18**

(i.e., remainders of the trenches **15**) are formed among the conductive lines **1911**. The capping layer **192** is formed from the dielectric layer **17**. Each of the dielectric liners **193** is disposed between two corresponding ones of the conductive lines **1911**, and conformally covers a lateral wall of each of the two corresponding ones of the conductive lines **1911** and a portion of a top surface of the ILD layer **11**, such that each of the air gaps **18** is defined by cooperation of the capping layer **192** with a corresponding one of the dielectric liners **193**. Since the air gaps **18** are formed among the conductive lines **1911** to isolate the conductive lines **1911**, the line-to-line capacitance and the resistance-capacitance (RC) time delay may be reduced. In some embodiments, the conductive lines **1911** have a width (W) ranging from about 1 nm to about 1000 nm, a height (H) ranging from about 1 nm to about 1000 nm, and an aspect ratio of the height (H) to the width (W) of greater than about 1:1. If the height of conductive lines **1911** made of the topological material is less than 1 nm, the topological effect on the conduction of the conductive lines **1911** may vanish. If the height of conductive lines **1911** made of the topological material is greater than 1000 nm, the resistance of the conductive lines **1911** may not be competitive compared to that of conductive lines made of metal (e.g., copper (Cu)) with the same height. In some embodiments, the aspect ratio may range from about 3:2 to about 2:1. In some embodiments, the aspect ratio may be even greater than about 2:1.

(19) Referring to FIG. **1** and the example illustrated in FIG. **7**, the method **100** then proceeds to step **106**, where at least one etch stop layer and a dielectric layer are formed sequentially on the first interconnect layer. At least one etch stop layer **20** is formed on the first interconnect layer **19** by a suitable deposition process as is known in the art of semiconductor fabrication, such as PVD, CVD, ALD, PEALD, thermal ALD, PECVD, or the like. Other suitable deposition techniques are within the contemplated scope of the present disclosure. In some embodiments, the at least one etch stop layer **20** includes a stack assembly of at least one first etch stop layers **201** and at least one second etch stop layer **202** which are alternately stacked on the first interconnect layer **19**. In some embodiments, the number of the at least one first etch stop layer **201** is two and the number of the at least one second etch stop layer **202** is two. In some embodiments, the number of the at least one first etch stop layer **201** is one and the number of the at least one second etch stop layer **202** is one. The first etch stop layer **201** and the second etch stop layer **202** are different from each other, and each of them may independently include silicon nitride, silicon nitride doped with carbon, silicon oxide, silicon oxynitride, silicon oxynitride doped with carbon, amorphous carbon material, silicon carbide, silicon oxycarbide, other nitride materials, other carbide materials, aluminum oxide, other metal oxides, aluminum nitride, other metal nitrides (e.g., titanium nitride, or the like), boron nitride, boron carbide, and other low-k dielectric materials or low-k dielectric materials doped with one or more of carbon, nitrogen, and hydrogen, or other suitable materials. Other suitable materials are within the contemplated scope of the present disclosure. A dielectric layer **21** is then formed on the at least one etch stop layer **20**. The material and the process for forming the dielectric layer **21** may be the same as or similar to those for forming the ILD layer **11** described above with reference to FIG. **2**, and the details thereof are omitted for the sake of brevity.

(20) Referring to FIG. **1** and the examples illustrated in FIGS. **8** and **9**, the method **100** then proceeds to step **107**, where a second interconnect layer is formed. In some embodiments, a second interconnect layer **27** includes an interconnect structure **271**, which includes at least one via contact **2711** and at least one metal line **2712**. The at least one via contact **2711** and the at least one metal line **2712** are formed simultaneously using using a dual damascene processes.

(21) At least one trench **22** and at least one via opening **23** are formed by patterning the dielectric layer **21** and the at least one etch stop layer **20** using one or more etching processes (for example, but not limited to, a dry etching process, a wet etching process, or a combination thereof) through a pattern of opening formed in a patterned mask layer (not shown) so as to expose at least one of the conductive lines **1911** through the at least one via opening **23** and the at least one trench **22**, respectively. The at least one trench **22** is recessed downwardly from a top surface of the dielectric

layer **21**, and the at least one via opening **23** extends through the at least one etch stop layer **20** and is disposed below and in spatial communication with the at least one trench **22**, respectively, so as to form at least one integrated opening for the at least one of the conductive lines **1911** to be exposed therethrough, respectively.

(22) A metal barrier layer **24** is conformally formed in the at least one trench **22** and the at least one via opening **23** by a suitable deposition process as is known to those skilled in the art of semiconductor fabrication, for example, PVD, CVD, PECVD, ALD, PEALD, or the like. Other suitable deposition techniques are within the contemplated scope of the disclosure. The metal barrier layer **24** may include, for example, but not limited to, cobalt, ruthenium, tungsten, titanium nitride, zirconium oxide, aluminum oxide, yttrium oxide, aluminum oxynitride, hafnium oxide, hafnium zirconium oxide, hafnium silicon oxide, hafnium silicon oxynitride, zirconium silicon oxide, hafnium zirconium silicon oxide, hafnium aluminum oxide, hafnium aluminum nitride, zirconium aluminum oxide, ytterbium oxide, tantalum, tantalum nitride, tantalum silicon nitride, tantalum oxide, tantalum silicon oxide, titanium nitride, titanium silicon nitride, titanium oxide, titanium silicon oxide, or combinations thereof. Other suitable metal barrier materials are within the contemplated scope of the present disclosure.

(23) A metal liner layer **25** is then conformally formed on the metal barrier layer **24** by a suitable deposition process as is known to those skilled in the art of semiconductor fabrication, for example, PVD, CVD, PECVD, ALD, PEALD, or the like. Other suitable deposition techniques are within the contemplated scope of the present disclosure. In some embodiments, the metal liner layer **25** may include, for example, but not limited to, metals (e.g., copper (Cu), silver (Ag), gold (Au), aluminum (Al), nickel (Ni), cobalt (Co), ruthenium (Ru), iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), tungsten (W), molybdenum (Mo), tantalum (Ta), or the like), alloys of the metals, the nitride, carbide, silicide compounds of the metals, or the like, or combinations thereof. Other suitable metal liner materials are within the contemplated scope of the present disclosure. In some embodiments, the metal liner layer **25** may have a thickness ranging from about 1 nm to about 10 nm. If the thickness is less than 1 nm, the liner properties of the metal liner layer **25** may be degraded. If the thickness is greater than 10 nm, the volume for filling a metal material to form an interconnect structure is undesirably decreased such that the conduction volume of the interconnect structure thus formed is decreased and the resistance of the interconnect structure is increased.

(24) A metal material **26** is filled into the at least one trench **22** and the at least one via opening **23** by a suitable deposition process as is known to those skilled in the art of semiconductor fabrication, for example, selective or non-selective PVD, selective or non-selective CVD, selective or non-selective PECVD, selective or non-selective ALD, selective or non-selective PEALD, electroless deposition (ELD), electro-chemical plating (ECP), or the like, and a planarization treatment (e.g., CMP) is then performed to remove excess of the metal material **26** over the dielectric layer **21** so as to form the second interconnect layer **27** including the interconnect structure **271**. The interconnect structure **271** includes the at least one via contact **2711** and the at least one metal line **2712**. The at least one via contact **2711** is disposed in a lower portion of the second interconnect layer **27**. The at least one metal line **2712** is disposed in an upper portion of the second interconnect layer **27**, and is electrically connected to at least one of the conductive lines **1911** through the at least one via contact **2711**, respectively.

(25) In some embodiments, the metal material **26** may include, for example, but not limited to, metals (e.g., Cu, Ag, Au, Al, Ni, Co, Ru, Ir, Pt, Pd, Os, W, Mo, Ta, or the like), alloys thereof possessing promising conductive properties, or the like. Other suitable metal materials are within the contemplated scope of the present disclosure. The deposition process for forming the interconnect structure **271** may be performed at a temperature ranging from about 25° C. to about 1000° C. If the temperature of the deposition process is higher than 1000° C., the materials and the components disposed proximate to the interconnect structure **271** may be damaged.

(26) Referring to the examples illustrated in FIGS. 10 to 12, in some alternative embodiments, the interconnect structure 271 includes the at least one via contact 2711 and at least one conductive line 2712'. The at least one via contact 2711 is disposed in a lower portion of the second interconnect layer 27. The at least one conductive line 2712' is disposed in an upper portion of the second interconnect layer 27, and is electrically connected to at least one of the conductive lines 1911 through the at least one via contact 2711, respectively. The at least one via contact 2711 and the at least one conductive line 2712' are separately formed using two single damascene processes.

(27) The at least one via opening 23 is formed by patterning the dielectric layer 21 and the at least one etch stop layer 20 using one or more etching processes (for example, a dry etching process, a wet etching process, or a combination thereof) through a pattern of opening formed in a patterned mask layer (not shown), so as to expose at least one of the conductive lines 1911 through the at least one via opening 23, respectively.

(28) The metal barrier layer 24 is conformally formed in the at least one via opening 23, and the metal liner layer 25 is then conformally formed on the metal barrier layer 24. The materials and the processes for forming the metal barrier layer 24 and the metal liner layer 25 may be the same as or similar to those described above with reference to FIG. 8, and the details thereof are omitted for the sake of brevity.

(29) The metal material 26 is filled into the at least one via opening 23, and a planarization treatment (e.g., CMP) is then performed to remove a portion of the metal material 26 and optionally a portion of the dielectric layer 21, a portion of the metal barrier layer 24, and a portion of the metal liner 25 so as to form the at least one via contact 2711. Examples of the metal material 26 and the process for depositing the metal material 26 may be the same as or similar to those described above with reference to FIG. 9, and the details thereof are omitted for the sake of brevity.

(30) Formation of the at least one conductive line 2712' may be the same as or similar to those described above with reference to FIGS. 2 to 6, and the details thereof are omitted for the sake of brevity.

(31) In some alternative embodiments, the metal material 26 is directly filled into the at least one via opening 23 without formation of the metal barrier layer 24 and the metal liner layer 25.

(32) Referring to the example illustrated in FIG. 13, in some alternative embodiments, the capping layer 192 is completely filled into the spaces defined by the dielectric liners 193 by completely filling the dielectric layer 17 into the trenches 15 during the stage illustrated in FIGS. 4 and 5, such that the first interconnect layer 19 is not formed with air gaps among the conductive lines 1911.

(33) Referring to the example illustrated in FIG. 14, in some alternative embodiments, the conductive lines 1911 have an aspect ratio (a ratio of height (H) to width (W)) of less than about 1:1 so as to reduce line-to-line capacitance. In some embodiments, the aspect ratio may range from about 1:2 to about 2:3. In some embodiments, the aspect ratio may be even less than about 1:2.

(34) Referring to the example illustrated in FIG. 15, in some alternative embodiments, the metal barrier layer 24 is conformally formed in the at least one trench 22 and the at least one via opening 23 by a suitable selective deposition process as is known to those skilled in the art of semiconductor fabrication, for example, selective PVD, selective CVD, selective ALD, selective ELD, or the like, such that an opening is formed at a bottom of the metal barrier layer 24. Other suitable deposition techniques are within the contemplated scope of the present disclosure. Since the at least one via contact 2711 of the interconnect structure 271 thus formed does not include the metal barrier layer 24 at a bottom thereof, the contact resistance between the at least one via contact 2711 and at least one of the conductive lines 1911 may be reduced.

(35) Referring to the example illustrated in FIG. 16, in some alternative embodiments, the interconnect structure 271 includes the at least one via contact 2711 and the at least one conductive line 2712' electrically connected to at least one of the conductive lines 1911 through the at least one via contact 2711, respectively. The at least one via contact 2711 and the at least one conductive line 2712' may be separately formed using two single damascene processes. The at least one via contact

2711 includes the topological material, and is laterally covered by the dielectric liner **193**. The dielectric liner **193** is formed by a suitable selective deposition process as is known to those skilled in the art of semiconductor fabrication, for example, selective PVD, selective CVD, selective ALD, selective ELD, or the like, such that an opening is formed at a bottom of the dielectric liner **193**, and such that the at least one via contact **2711** made of the topological material is electrically connected to at least one of the conductive lines **1911**, respectively. Other suitable deposition techniques are within the contemplated scope of the present disclosure.

(36) In the semiconductor device of the present disclosure, the topological material is applied to form an interconnect structure based on the exotic transport property thereof. The interconnect structure in a three-dimensional configuration including the topological material possesses an exotic surface state, which provides a two-dimensional like conduction similar to the conduction of graphene. In addition, the topological material has an exotic three-dimensional conduction nature, such that the interconnect structure in the three-dimensional configuration including the topological material does not suffer a high contact resistance, unlike the contact resistance of a two-dimensional material (e.g., graphene) resulting from a slow out-of-plan transport property. The interconnect structure including the topological material has a low contact resistance to metal lines and/or via contacts. Therefore, when an interconnect structure is scaled down, the interconnect structure including the topological material may have a relatively low line resistance compared with that of the interconnect structure made of metal (e.g., copper). Furthermore, when the interconnect structure including the topological material is integrated with air gaps formed among the conductive lines of the interconnect structure, the line-to-line capacitance of the interconnect structure may be reduced and the RC time delay may be reduced accordingly. Moreover, when the aspect ratio (a ratio of height to width) of the conductive lines of the interconnect structure including the topological material is lowered, the line-to-line capacitance of the interconnect structure may also be reduced and the RC time delay may be reduced accordingly.

(37) In accordance with some embodiments of the present disclosure, a semiconductor device includes a substrate and an interconnect layer disposed over the substrate. The interconnect layer includes an interconnect structure which includes a topological material.

(38) In accordance with some embodiments of the present disclosure, the topological material includes a topological insulator, a topological semimetal, or a combination thereof.

(39) In accordance with some embodiments of the present disclosure, the interconnect structure further includes a magnetic material.

(40) In accordance with some embodiments of the present disclosure, the interconnect structure includes a via contact disposed in a lower portion of the interconnect layer, and a conductive line disposed in an upper portion of the interconnect layer and electrically connected to the via contact. At least one of the via contact and the conductive line includes the topological material.

(41) In accordance with some embodiments of the present disclosure, both the via contact and the conductive line include the topological material.

(42) In accordance with some embodiments of the present disclosure, the interconnect layer further includes a dielectric liner laterally covering the via contact.

(43) In accordance with some embodiments of the present disclosure, the interconnect structure includes a first conductive line and a second conductive line which are spaced apart from each other to form a trench therebetween and which include the topological material. The interconnect layer further includes a dielectric liner which is disposed in the trench and between the first and second conductive lines and which conformally covers a lateral wall of each of the first and second conductive lines, and a capping layer disposed in the trench.

(44) In accordance with some embodiments of the present disclosure, the dielectric liner cooperates with the capping layer to define an air gap.

(45) In accordance with some embodiments of the present disclosure, the conductive lines have an aspect ratio of height to width of less than 1:1.

(46) In accordance with some embodiments of the present disclosure, the conductive lines have an aspect ratio of height to width of less than 1:1.

(47) In accordance with some embodiments of the present disclosure, a semiconductor device includes a substrate, a first interconnect layer, and a second interconnect layer. The first interconnect layer is disposed over the substrate, and includes a first interconnect structure containing a first topological material. The second interconnect layer is disposed on the first interconnect layer, and includes a second interconnect structure which includes a via contact disposed in a lower portion of the second interconnect layer and a conductive line disposed in an upper portion of the second interconnect layer and electrically connected to the first interconnect structure through the via contact. At least one of the via contact and the conductive line include a second topological material.

(48) In accordance with some embodiments of the present disclosure, each of the first and second topological materials independently includes a topological insulator, a topological semimetal, or a combination thereof.

(49) In accordance with some embodiments of the present disclosure, both the via contact and the conductive line include the second topological material.

(50) In accordance with some embodiments of the present disclosure, the first interconnect structure includes a first conductive line and a second conductive line which are spaced apart from each other to form a first trench therebetween and one of which is electrically connected to the conductive line of the second interconnect structure through the via contact. The first interconnect layer further includes a dielectric liner which is disposed in the first trench and between the first and second conductive lines and which conformally covers a lateral wall of each of the first and second conductive lines, and a capping layer disposed in the first trench.

(51) In accordance with some embodiments of the present disclosure, the dielectric liner cooperates with the capping layer to define an air gap.

(52) In accordance with some embodiments of the present disclosure, the second interconnect structure includes two of the conductive lines which are spaced apart from each other to form a second trench therebetween, and one of which is electrically connected to a corresponding one of the first and second conductive lines of the first interconnect structure through the via contact. The second interconnect layer further includes a first dielectric liner, a second dielectric liner, and a capping layer. The first dielectric liner laterally covers the via contact. The second dielectric liner is disposed in the second trench and between the two conductive lines of the second interconnect structure, and conformally covers a lateral wall of each of the two conductive lines of the second interconnect structure. The capping layer is disposed in the second trench.

(53) In accordance with some embodiments of the present disclosure, a method for manufacturing a semiconductor device includes forming a topological material layer over a substrate; patterning the topological material layer to form a trench in the topological material layer; conformally forming a dielectric liner in the trench; and filling a capping layer into the trench to form a first interconnect layer including a first interconnect structure formed from the topological material layer.

(54) In accordance with some embodiments of the present disclosure, the method for manufacturing a semiconductor device further includes forming at least one etch stop layer on the first interconnect layer; forming a dielectric layer on the at least one etch stop layer; and forming a second interconnect structure which extends through the dielectric layer and the at least one etch stop layer and which includes a via contact and a conductive line electrically connected to the first interconnect structure through the via contact, at least one of the via contact and the conductive line including a topological material.

(55) In accordance with some embodiments of the present disclosure, one of the via contact and the conductive line is made of the topological material, and the other one of the via contact and the conductive line is made of a metal material. The via contact and the conductive line are formed separately using two single damascene processes.

(56) In accordance with some embodiments of the present disclosure, both the via contact and the conductive line are made of the topological material. The via contact and the conductive line are formed simultaneously using a dual damascene processes.

(57) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes or structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device comprising: a substrate; and an interconnect layer disposed over the substrate and including an interconnect structure, the interconnect structure including a topological material, wherein the interconnect structure includes a first conductive line and a second conductive line which are spaced apart from each other to form a trench between the first conductive line and the second conductive line and which include the topological material; the interconnect layer further includes a first dielectric liner which is disposed in the trench and between the first conductive line and the second conductive line and which conformally covers a lateral wall of each of the first conductive line and the second conductive line, and a capping layer disposed in the trench; and the first dielectric liner cooperates with the capping layer to define an air gap.
2. The semiconductor device according to claim 1, wherein the topological material includes a topological insulator, a topological semimetal, or a combination thereof.
3. The semiconductor device according to claim 1, wherein the interconnect structure further includes a magnetic material.
4. The semiconductor device according to claim 1, wherein the interconnect structure further includes: a via contact disposed on a corresponding one of the first conductive line and the second conductive line, and a conductive line disposed on and electrically connected to the via contact, at least one of the via contact and the conductive line including the topological material.
5. The semiconductor device according to claim 4, wherein both the via contact and the conductive line include the topological material.
6. The semiconductor device according to claim 5, wherein the interconnect layer further includes a second dielectric liner laterally covering the via contact.
7. The semiconductor device according to claim 1, wherein each of the first conductive line and the second conductive line has an aspect ratio of height to width of greater than 1:1.
8. The semiconductor device according to claim 1, wherein each of the first conductive line and the second conductive line has an aspect ratio of height to width of less than 1:1.
9. A semiconductor device comprising: a substrate; a first interconnect layer disposed over the substrate, and including a first interconnect structure containing a first topological material; and a second interconnect layer disposed on the first interconnect layer, and including a second interconnect structure which includes a via contact disposed in an lower portion of the second interconnect layer and a first conductive line disposed in an upper portion of the second interconnect layer and electrically connected to the first interconnect structure through the via contact, at least one of the via contact and the first conductive line including a second topological material.
10. The semiconductor device according to claim 9, wherein each of the first topological material and the second topological material independently includes a topological insulator, a topological

semimetal, or a combination thereof.

11. The semiconductor device according to claim 9, wherein both the via contact and the first conductive line include the second topological material.

12. The semiconductor device according to claim 11, wherein the first interconnect structure includes a first conductive line and a second conductive line which are spaced apart from each other to form a first trench between the first conductive line and the second conductive line of the first interconnect structure and one of which is electrically connected to the first conductive line of the second interconnect structure through the via contact; and the first interconnect layer further includes a dielectric liner which is disposed in the first trench and between the first conductive line and the second conductive line of the first interconnect structure and which conformally covers a lateral wall of each of the first conductive line and the second conductive line of the first interconnect structure, and a capping layer disposed in the first trench.

13. The semiconductor device according to claim 12, wherein the dielectric liner cooperates with the capping layer to define an air gap.

14. The semiconductor device according to claim 12, wherein the second interconnect structure further includes a second conductive line which is spaced apart from the first conductive line of the second interconnect structure to form a second trench between the first conductive line and the second conductive line of the second interconnect structure; and the second interconnect layer further includes a first dielectric liner laterally covering the via contact, a second dielectric liner which is disposed in the second trench and between the first conductive line and the second conductive line of the second interconnect structure and which conformally covers a lateral wall of each of the first conductive line and the second conductive line of the second interconnect structure, and a capping layer disposed in the second trench.

15. A semiconductor device comprising: a substrate; a dielectric layer disposed over the substrate; and an interconnect layer disposed on the dielectric layer, and including an interconnect structure and a dielectric liner in contact with the dielectric layer and laterally covering the interconnect structure, the interconnect structure including a topological material which includes a topological insulator, a topological semimetal, or a combination thereof, wherein the interconnect layer further includes a capping layer cooperating with the dielectric liner to define an air gap.

16. The semiconductor device according to claim 15, wherein the topological insulator includes bismuth telluride, antimony telluride, bismuth-antimony alloy, bismuth selenide, or combinations thereof.

17. The semiconductor device according to claim 15, wherein the topological semimetal includes niobium phosphide, tantalum arsenide, niobium arsenide, tantalum phosphide, tungsten ditelluride, molybdenum ditelluride, tungsten diphosphide, molybdenum diphosphide, cadmium arsenide, platinum stannide, lanthanum antimonide, lanthanum bismuthide, platinum bismuthide), zirconium pentatelluride, hafnium pentatelluride, lead tantalum selenide, zirconium silicon sulfide, hafnium silicide sulfide, niobium arsenide, tantalum arsenide, or combinations thereof.

18. The semiconductor device according to claim 15, wherein the interconnect structure further includes a magnetic material, which includes cobalt, iron, nickel, or combinations thereof.

19. The semiconductor device according to claim 15, wherein the dielectric liner includes a base portion in contact with the dielectric layer, and a wall portion extending from the base portion and laterally covering the interconnect structure.

20. The semiconductor device according to claim 19, wherein the capping layer is in contact with an upper part of the wall portion of the dielectric liner.
